

- 1 A pair of digital calipers measures the external diameter of a metal pipe to be 2.780 cm. The internal diameter of the pipe is 2.020 cm. What is the thickness of the wall of the metal pipe?

A 0.280 cm **B** 0.380 cm **C** 0.560 cm **D** 0.760 cm

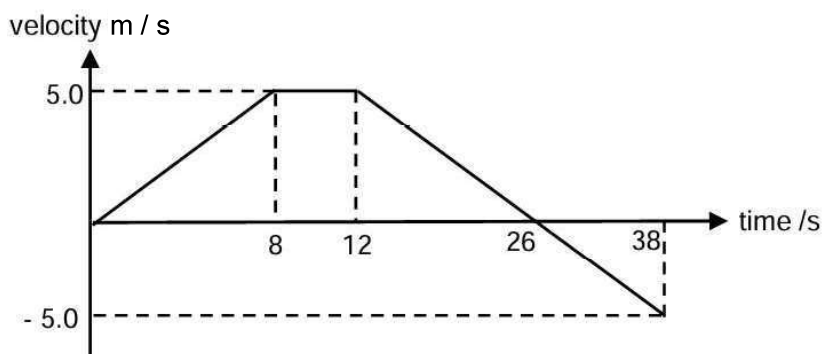
- 2 The diameter of four spheres W, X, Y and Z are shown in the table below.

object	diameter
W	0.002 mm
X	40 nm
Y	0.00005 Tm
Z	0.25 μ m

Which of the following correctly lists the size of objects in ascending order?

- A** W, X, Y, Z
B X, Z, W, Y
C Z, X, W, Y
D X, Z, Y, W

- 3 The diagram shows the velocity-time graph of a cyclist's journey.



What is the average speed for the cyclist's entire journey?

A 1.0 m / s **B** 1.2 m / s **C** 1.8 m / s **D** 2.8 m / s

- 4 A feather falls through a vacuum. Which row describes the velocity and acceleration of the object?

	velocity	acceleration
A	constant	constant
B	constant	increasing
C	increasing	constant
D	increasing	increasing

- 5 Two forces act on a stationary object. The magnitudes of the forces are 5.0 N and 12.0 N respectively. Which could be the magnitude of the resultant force acting on the object?

A 6.0 N **B** 8.5 N **C** 17.2 N **D** 19.0 N